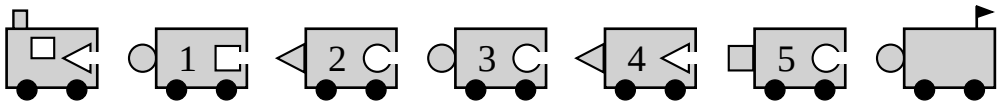
Math Kangaroo Practice

Grade 3-4 · 3 questions · harder · seed 43042

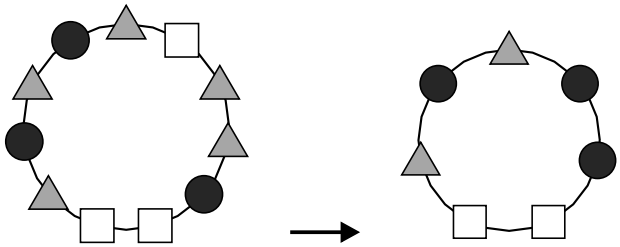
Part B: Each question is worth 4 points

1. Emma has these pieces: an engine, a last wagon and 5 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. How many different trains can Emma build with **exactly 3** numbered wagons **between** the engine and the last wagon?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 10

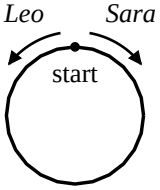
2. Mia takes away 4 neighbouring stones from the bracelet on the left. Then the bracelet looks like the one on the right. Which stones did Mia take away?



A. ●△△△ ***B.*** ●□△△ ***C.*** △△△△ ***D.*** □△△△ ***E.*** □□△△

Part C: Each question is worth 5 points

3. Sara and Leo start together at the same spot of a round track. Sara runs clockwise and Leo runs anticlockwise. Sara needs 12 minutes to run once around, Leo only needs 6 minutes. After 48 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



A. 2 ***B.*** 8 ***C.*** 11 ***D.*** 12 ***E.*** 13

Answer Key

1. **B**	**2. D**	**3. D**

Solutions

1. B (2) — Count only the numbered wagons, not the engine and the last wagon. Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–2–1–5–T, H–4–2–3–T. That is 2.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

2. D — Count each shape before and after: circles 3 → 3, squares 3 → 2, triangles 5 → 2. So Mia took away 1 square and 3 triangles. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · which_removed]

3. D (12) — Between two meetings they run once around the track together. In 48 minutes Sara runs 4 and Leo runs 8 times around: 4 + 8 = 12 laps together, so 12 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]